SRI CHAITANYA EDUCATIONAL INSTITUTIONS,INDIA.

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI,CHANDIGARH

SEC : SR AIIMS S60 **DATE :14-12-24**

SUB :BOTANY **NEET PRE-WEEKEND TEST-09** **Max.Marks : 360**

IMPORTANT INSTRUCTIONS :

- Pattern of the Entrance Examination:-
  Paper containing 180 objective type questions ,from Biology, Physics and Chemistry
- Use Blue/Black Ball Point Pen only to darken the appropriate circle. Answers marked with pencil would not be evaluated.
- Each item carries 4marks. For each correct response the candidate will get 4 marks. For each incorrect response 1mark will be deducted from the total score.

01. The term 'Nuclein' for the genetic material was used by :
(1) Mendel (2) Franklin
(3) Meischer (4) Chargaff

02. Which one of the following does not follow the central dogma of molecular biology ?
(1) Pea (2) *Mucor*
(3) *Chlamydomonas* (4) HIV

03. A segment of DNA has 120 adenine and 120 cytosine bases. The total number of nucleotides present in the segment is :
(1) 120 (2) 240
(3) 60 (4) 480

04. Match List I with List II

List – I	List – II
A. Frederick Griffith	I. Genetic code
B. Francis Jacob and Jacque Monod	II. Semi – conservative mode of DNA
C. Har Gobind Khorana	III. Transformation
D. Meselson and Stahl	IV. *Lac* operon

Choose the correct answer from the options given below :
(1) A – I , B – III, C – IV, D – I
(2) A – IV , B – I, C – II, D – III
(3) A – III , B – II, C – I, D – IV
(4) A – III , B – IV, C – I, D – II

05. Ten E.coli cells $^{15}N-dsDNA$ are incubated in medium containing ^{14}N nucleotide. After 60 minutes, how many E.coli cells will have DNA totally free from ^{15}N ?
(1) 60 cells (2) 80 cells
(3) 20 cells (4) 40 cells

06. Which of the following RNAs is not required for the synthesis of protein ?
(1) SiRNA (2) mRNA
(3) tRNA (4) rRNA

07. During DNA replication, the strands separate by :
(1) DNA polymerase
(2) Topoisomerase
(3) Unwindase/helicase
(4) SSB

08. Which is "Only enzyme" that has "Capability" to catalyse initiation, Elongation and Termination in the process of transcription in prokaryotes ?
(1) DNase
(2) DNA dependent DNA polymerase
(3) DNA dependent RNA polymerase
(4) DNA Ligase

09. Name the enzyme that facilitates opening of DNA helix during transcription :
(1) DNA helicase
(2) DNA polymerase
(3) RNA polymerase
(4) DNA ligase

10. What initiation and termination factors are involved in transcription in Eukaryotes ?
(1) α and σ, respectively
(2) σ and ρ, respectively
(3) α and β, respectively
(4) β and γ, respectively

11. Removal of RNA polymerase III from nucleoplasm will affect the synthesis of :
(1) mRNA (2) rRNA
(3) tRNA (4) hnRNA

12. Which one of the following makes use to RNA as a template to synthesize DNA ?
(1) Reverse transcriptase
(2) DNA dependant RNA polymerase
(3) DNA polymerase
(4) RNA polymerase

13. In the genetic code dictionary, how many codons are used to code for all the 20 amino acids ?
(1) 64 (2) 61
(3) 60 (4) 20

14. Out of 64 codons, 61 codons code for 20 type of amino acids; it is called :
(1) Degeneracy of genetic code
(2) Unambiuguous nature of genetic code
(3) Wobbling of codon
(4) Universality of codon

15. Under which of the following conditions will there be no change in the reading frame of following mRNA ?
5' *AACAGCGGUGCUAUU* 3' :
(1) Insertion of G at 5th position
(2) Deletion of G from 5th position
(3) Insertion of A and G at 4th and 5th positions respectively
(4) Deletion of GGU from 7th, 8th and 9th positions

16. Select the correct options :

Direction of RNA synthesis	Direction of reading the template DNA strand
(1) 3'–5'	5'–3'
(2) 5'–3'	5'–3'
(3) 3'–5'	3'–5'
(4) 5'–3'	3'–5'

17. mRNA is synthesized on DNA template in which of the following direction ?
(1) $5' \rightarrow 3'$ (2) $3' \rightarrow 5'$
(3) Both direction (4) Any direction

18. Match the following genes of the Lac operon with their respective products :

(A) i gene	(i) $\beta-$ galactosidase
(B) z gene	(ii) Permease
(C) a gene	(iii) Repressor
(D) y gene	(iv) Transacetylase

Select the correction option :

	A	B	C	D
(1)	i	iii	ii	iv
(2)	iii	i	ii	iv
(3)	iii	i	iv	ii
(4)	iii	iv	i	ii

19. Which of the following is required as inducer(s) for the expression of *Lac* operon ?
(1) Glucose
(2) Galactose
(3) Lactose
(4) Lactose and galactose

20. Which one of the following is wrongly matched ?
(1) Repressor protein – Binds to operator to stop enzyme synthesis
(2) Operon – Structural genes, operator and promoter
(3) Transcription – Writing information from DNA to tRNA
(4) Translation – Using information in mRNA to make protein

21. Select the two correct statements out of the four (A – D) given below about *Lac* operon.
(A) Glucose or galactose may bind with the repressor and inactivate it
(B) In the absence of lactose the repressor binds with the operator region
(C) The Z – gene codes for permease
(D) This was elucidated by Francis Jacob and Jacque Monod
The correct statements are :
(1) B and C (2) A and C
(3) B and D (4) A and B

22. Assertion : DNA is considered to be better genetic material than RNA for most organisms.
Reason : 2'–*OH* group present in DNA makes it labile and less reactive.
Directions : In the following questions, a statement of assertion is followed by a statement of reason.
Mark the correct choice as :
(1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.

ZOOLOGY

23. How many of the following statements correctly describe the spleen?
(A) It is the largest lymphatic organ in the body

(B) It traps lymph-borne antigens and presents them to lymphocytes.
(C) It provides a microenvironment for the maturation of B-cells and T-cells
(D) It acts as a large reservoir of erythrocytes
(1) A and B (2) A, B, and C
(3) A and D (4) A, B, C, and D

24. Which of the following statements is incorrect?
(1) The lymph nodes are secondary lymphoid organs.
(2) MALT constitutes 50% of the lymphoid tissue in the human body.
(3) The thymus is quite large at birth but decreases in size with age.
(4) The bone marrow and thymus provide sites for the interaction of lymphocytes with antigens.

25. Match the following

COLUMN I	COLUMN II
(A) Thymus gland	i) Anti retro-viral drugs
(B) Treatment of AIDS	ii) Filter of blood
(C) Spleen	iii) Screening test for AIDS
(D) ELISA	iv) Beneath thebreast bone

(1) A-i, B-iii, C-iv, D-ii
(2) A-iv, B-i, C-ii, D-iii
(3) A-iv, B-iii, C-ii, D-I
(4) A-iv, B-ii, C-iii, D-i

26. HIV that causes AIDS, first starts destroying
(1) Thrombocytes
(2) B-lymphocytes
(3) Leucocytes
(4) Helper T-lymphocytes

27. Assertion: Alcohol consumption can impair the functioning of the liver, leading to conditions like fatty liver, cirrhosis.
Reason : Alcohol is metabolized by the liver, and excessive consumption leads to the accumulation of toxic byproducts, which damage liver cells and disrupt their normal function
(1) Both assertion and reasoning are correct, and the reasoning correctly explains the assertion
(2) Both assertion and reasoning are correct, but the reasoning does not correctly explain the assertion
(3) The assertion is correct, but the reasoning is incorrect
(4) Both assertion and reasoning are incorrect

28. Statement 1: Cancer cells have lost the contact inhibition property.
Statement 2: Metastasis is the most feared property of cancer cells.
(1) Both statement 1 and statement 2 are incorrect
(2) Both statement 1 and statement 2 are correct
(3) Statement 1 is correct, but statement 2 is incorrect
(4) Statement 1 is incorrect, but statement 2 is correct

29. Which diagnostic technique uses a strong magnetic field and non-ionizing radiation to detect physiological changes in living tissues?
(1) Computed Tomography (CT)
(2) X-rays
(3) Magnetic Resonance Imaging (MRI)
(4) Mammography

30. Cancer patients are given biological response modifiers like ___ to activate the immune system.
(1) γ-globulins (2) β-interferons
(3) α-interferons (4) Cyclosporine

31. Match the following columns and select the correct option:

Column I	Column II
a. Heroin	i. Cannabinoid
b. Hashish	ii. Cirrhosis
c. Cocaine	iii. Diacetylmorphine
d. Alcohol	iv. Euphoria

	a	b	c	d
(1)	i	iii	iv	ii
(2)	iii	i	iv	ii
(3)	iii	i	ii	iv
(4)	iv	ii	iii	i

32. Which of the following is a primary effect of cocaine use?
(1) It has a potent stimulating action on central nervous system
(2) It causes long-term memory loss and cognitive impairment.
(3) It primarily functions as a sedative by inhibiting neurotransmitter release
(4) It is a direct cause of liver failure after chronic use.

33. Which of the following substances found in tobacco smoke reduces the amount of oxygen bound to hemoglobin?
(1) Nicotine
(2) Carbon monoxide
(3) Carbon dioxide
(4) Benzopyrene

34. Currently, about (A) recombinant therapeutics have been approved for human use worldwide, and (B) of these are being marketed in India.
Select the correct option for A and B:
(1) A - 30, B - 5
(2) A - 30, B - 12
(3) A - 57, B - 30
(4) A - 45, B – 15

35. Why is human insulin preferred over animal insulin for the treatment of diabetes mellitus?
(1) Animal insulin is digested in the human alimentary canal before it is absorbed
(2) Human insulin is water-soluble, whereas animal insulin is lipid-soluble
(3) Animal insulin may cause allergic reactions in some patients
(4) Animal insulin acts as a hyperglycaemic hormone, unlike human insulin

36. Why can't insulin be administered orally?
(1) It can elicit an immune response in the human body
(2) It is synthesized as human proinsulin and requires processing to become functional
(3) It will be digested in the alimentary canal before it is absorbed.
(4) It is a peptide hormone and can bind only to intracellular receptors

37. Which of the following is a diagnostic test based on the principle of antigen-antibody interaction?
(1) PCR (2) ECG
(3) ELISA (4) MRI

38. Which of the following techniques is primarily used for detecting specific DNA sequences in molecular diagnosis?
(1) PCR (Polymerase Chain Reaction)
(2) ELISA (Enzyme-Linked Immunosorbent Assay)
(3) Western blotting
(4) X-ray diffraction

39. The side effects of the use of anabolic steroids in males includes :
(1) Acne, increased aggressiveness, premature baldness
(2) Mood swings, depression
(3) Reduction in size of the testicles, decreased sperm production, breast enlargement, enlargement of prostate gland
(4) All are correct

40. In gene therapy for treating adenosine deaminase deficiency disorder, which of the patient's blood cells are used?
(1) Erythrocytes
(2) Lymphocytes
(3) Both erythrocytes and lymphocytes
(4) Thrombocyte

41. The vector used in the first clinical gene therapy was
(1) retrovirus (2) adenovirus
(3) flavivirus (4) herpesvirus

42. Which of the following could be a permanent cure for ADA deficiency in a patient?
(1) Bone marrow transplantation
(2) Injection of ADA-corrected lymphocytes into the patient
(3) Enzyme replacement therapy.
(4) Gene isolated from marrow cells producing ADA is introduced into cells at early embryonic stages.

43. Today, transgenic models exist for many human diseases, such as:
(i) Cancer
(ii) Cystic fibrosis
(iii) Rheumatoid arthritis
(iv) Alzheimer's disease
(1) (i) and (iii) only
(2) (ii) and (iii) only
(3) (i), (ii), and (iii) only
(4) (i), (ii), (iii), and (iv)

44. Match the following

Column – I	Column – II
A) Alpha-1-antitrypsin.	I) Transgenic mice
B) Alpha – lactalbumin	II) Normal physiology and development
C) Polio vaccine	III) Transgenic cow Rosie
D) Insulin growth factor	IV) Treat emphysema

A B C D
(1) IV III II I
(2) IV III I II

(3) I III II IV
(4) I III IV II

45. Transgenic animals are useful for how many of the following purposes?
(i) Vaccine safety testing
(ii) Studying diseases
(iii) Production of useful biological products
(iv) Chemical safety testing
(1) 4 (2) 3
(3) 2 (4) 1

PHYSICS

46. The radius of both deuterium and tritium are approximately 2.0 fm. The temperature to which the gas must be heated to initiate the fusion reaction is
(1) $3\times10^{9}K$ (2) $2\times10^{9}K$
(3) $3\times10^{12}K$ (4) $2\times10^{6}K$

47. The threshold wavelength for photo electric emission from a material is 5000 A^{0}. Photo electrons will be emitted when the material is illuminated with monochromatic radiation from
(1) 100 watt infrared lamp
(2) 1 watt infrated lamp
(3) 50 watt sodium vapour lamp
(4) 50 watt ultra violet lamp

48. When a mono chromatic point source of light is at a distance of 0.2 m from a photo electric cell, the cut off voltage and saturation current are respectively 0.6V and 18 mA. If the same source is placed 0.6 m away from the photo electric cell then
(1) the stopping potential will be 0.6V
(2) the saturation current will be 2 mA
(3) neither 1 nor 2
(4) both 1 and 2

49. V_1 and V_2 are the stopping potentials for the incident radiations of wave lengths λ_1 and λ_2 respectively are incident on a metallic surface. If $\lambda_1 = 3\lambda_2$
(1) $V_1 > 3V_2$ (2) $V_2 = 3V_1$
(3) $V_1 > \frac{V_2}{3}$ (4) $V_1 < \frac{V_2}{3}$

50. A stationary shell of mass '5m' explodes in to two parts and their masses are in the ratio 2 : 3, then the ratio of their deBroglie wavelengths is
(1) 2 : 3 (2) 3 : 2
(3) 9 : 4 (4) 1 : 1

51. If the velocity of the particle reduced to one third, then the percentage increases in its deBroglie wavelength is
(1) 100% (2) 200%
(3) 300% (4) 400%

52. A particle is projected horizontally with a velocity of 10m/s. What will be the ratio of deBroglie wavelength of particle when velocity vector makes an angle 45^{0} and 60^{0} with horizontal is
(1) 1 : 1 (2) $\sqrt{3}:1$
(3) $2:\sqrt{3}$ (4) $\sqrt{2}:1$

53. Ionization potential of hydrogen atom is 13.6 eV Hydrogen atoms in the ground state are excited by monochromatic ration of photon energy 12.1 eV. According to Bohr's theory, the spectral lines emitted by hydrogen will be
(1) One (2) Two
(3) Three (4) Four

54. The ratio of wavelengths of the last line of Balmer series and the last line of Lyman series is :
(1) 0.5 (2) 2
(3) 1 (4) 4

55. Bombardment of lithium with protons gives rise to the following reaction : $_3Li^{7} + {}_1H^{1} \rightarrow 2\left({}_2He^{4}\right) + Q$. Find the Q – value of the reaction. The atomic masses of lithium, proton and helium are 7.016u, 1.008 u and 4.004 u respectively.
(1) 12.904 Mev
(2) 14,904 Mev
(3) 10.904 Mev
(4) 16.904 Mev

56. A nuclear reactor has a power of 16 kW. If the energy released per fission is 200 MeV, the number of fissions per second is
(1) 5×10^{16} (2) 5×10^{17}
(3) 5×10^{14} (4) 5×10^{15}

57. Two long straight parallel conductors 10 cm apart, carry current of 5A each in opposite directions. Then the magnetic induction at a point mid way between them is
(1) $4\times10^{-5}T$ (2) $2\times10^{-5}T$
(3) $10^{-5}T$ (4) Zero

58. A 6Ω resistance is connected in the left gap of a meter bridge. In the second gap 3Ω and 6Ω are joined in parallel. The balance point of the bridge is at ____

(1) 75 cm (2) 60 cm
(3) 30 cm (4) 25 cm

59. Four resistance P,Q,R,X formed a wheat stone bridge. The bridge is balanced when $R = 100\Omega$. If P and Q are inter changed the bridge balance 121Ω for. The value of x is
(1) 100Ω (2) 200Ω
(3) 300Ω (4) 110Ω

60. The meter bridge can be used
(a) to determine an unknown resistance
(b) to compare two resistances
(c) to determine the specific resistance of the material of a wire.
(1) Only a, c are correct
(2) Only a, b are correct
(3) Only b, c are correct
(4) All a, b and c are correct

61. Which of the following acts as a circuit protection device ?
(1) Switch (2) Fuse
(3) Conductor (4) Inductor

62. If galvanometer and battery are interchanged in balanced Wheatstone bridge, then
(1) then battery discharges
(2) the bridge still balances
(3) the balance point is changed
(4) the galvanometer is damaged due to flow of

63. Two mutually perpendicular conductors carry currents i_1 and i_2 along x – axis and y – axis respectively. The locus of the points at which magnetic induction is zero.
(1) $x = \frac{i_1}{i_2} y$ (2) $y = \frac{i_1}{i_2} x$
(3) $y = \frac{i_2^2}{i_1} x$ (4) $x^2 + y^2 = \frac{i_1}{i_2}$

64. An ionized gas contains both positive and negative ions. If it is subjected simultaneously to an electric field along the +x direction and a magnetic field along the +z direction, then
(1) positive ions deflect towards +y direction and negative ions towards –y direction
(2) all ions deflect towards +y direction
(3) all ions deflect towards –y direction
(4) positive ions deflect towards –y direction and negative ions towards +y direction.

65. A long straight wire of radius 'a' carries a steady current i. The current is uniformly distributed across its cross section. The ratio of the magnetic fields at a/2 and 2a is
(1) 1 (2) 1/2
(3) 1/4 (4) 4

66. An electron accelerated through a potential difference 'V' passes through a uniform transverse magnetic field and experiences a force F. If the accelerating potential is increased to 2V, the electron in the same magnetic field will experience a force
(1) F (2) F/2
(3) $\sqrt{2}F$ (4) 2F

67. A magnetic pole of strength 8 Am is moved around a long straight wire carrying a current of 5A. The work done is
(1) $8\pi \times 10^{-7} J$ (2) $16\pi \times 10^{-6} J$
(3) $2\pi \times 10^{-7} J$ (4) zero

CHEMISTRY

68. The following are some statements about electrolytic cell
(A) in this, chemical energy is converted into electrical energy
(B) in this cell, electrons flow from cathode to anode
(C) in this, cell reduction takes place at cathode
(D) in this, cathode is a +ve electrode
The correct combination is
(1) only B (2) only C
(3) only C, D (4) only B, C

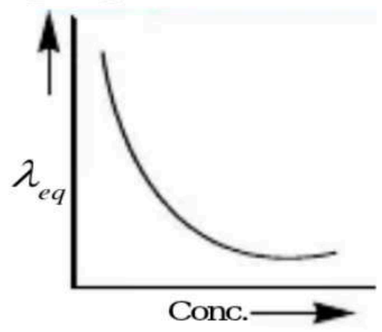
69. Which process occurs in the electrolysis of aqueous solution of nickel chloride at nickel anode ?
(1) $Ni^{2+} + 2e^- \rightarrow Ni$
(2) $2H^+ + 2e^- \rightarrow H_2$
(3) $2Cl^- \rightarrow Cl_2 + 2e^-$
(4) $Ni \rightarrow Ni^{2+} + 2e^-$

70. For a cell reaction involving a two – electron change, the standard e.m.f. of the cell is found to be 0.295 V at $25^0 C$. The equilibrium constant of the reaction at $25^0 C$ will be
(1) 1×10^{-10} (2) 29.5×10^{-2}
(3) 10 (4) 1×10^{10}

71. For the discharge of equal masses of the following ions, the number of electrons required is maximum in the case of
(1) H^+ (2) Cu^{2+}

(3) Ag^+ (4) Al^{3+}

72. A graph is drawn between the λ_{eq} values and concentration of an electrolyte. Which of the following electrolyte will correspond to the graph given ?



(1) KCl (2) $BaCl_2$

(3) H_2SO_4 (4) CH_3COOH

73. The correct order of equivalent conductance at infinite dilution of $LiCl, NaCl$ and KCl is

(1) $LiCl > NaCl > KCl$

(2) $KCl > NaCl > LiCl$

(3) $NaCl > KCl > LiCl$

(4) $LiCl > KCl > NaCl$

74. Which of the following statements is true for the electrochemical, Daniel cell ?

(1) Electrons flow from copper electrode to zinc electrode

(2) Current flows from zinc electrode to copper electrode

(3) Cations move towards copper electrode

(4) Cations move towards zinc electrode

75. For $I_2 + 2e \rightarrow 2I^-$, standard reduction potential = +0.54 volt. For $2Br^- \rightarrow Br_2 + 2e^-$, standard oxidation potential = +0.44 volt. Which of the following reactions is non – spontaneous ?

(1) $Br_2 + 2I^- \rightarrow 2Br^- + I_2$

(2) $Fe + Br_2 \rightarrow Fe^{2+} + 2Br^-$

(3) $Fe + I_2 \rightarrow Fe^{2+} + 2I^-$

(4) $I_2 + 2Br^- \rightarrow 2I^- + Br_2$

76. Given $E^0_{Cr+3/Cr} = -0.72V$; $E^0_{Fe^{2+}/Fe} = -0.42V$ The potential for the cell, $Cr\left|Cr^{3+}(0.1M \,\|\, Fe^{2+}(0.01M\right|Fe$ is

(1) -0.339V (2) -0.26V

(3) 0.26V (4) 0.339V

77. Consider the following four electrodes :

$A = Cu^{2+}(0.0001M)/Cu_{(s)}$

$B = Cu^{2+}(0.1M)/Cu_{(s)}$

$C = Cu^{2+}(0.01M)/Cu_{(s)}$

$D = Cu^{2+}(0.001M)/Cu_{(s)}$

If the standard reduction potential of Cu is +0.34V, the reduction potentials (in volts) of the above electrodes follow the order

(1) A >D >C >B

(2) B >C >D >A

(3) C >D >B >A

(4) A >B >C >D

78. The reaction occurring at cathode during the charging of lead storage batter is represented as

(1) $Pb \rightarrow Pb^{2+} + 2e^-$

(2) $Pb^{2+} + 2e^- \rightarrow Pb$

(3) $Pb^{2+} + SO_4^{2-} \rightarrow PbSO_4$

(4) $PbSO_4 + 2H_2O \rightarrow PbO_2 + 4H^+ + SO_4^{2-} + 2e^-$

79. The standard reduction potentials of three electrodes P, Q and R are respectively –1.76V, 0.34 V and 0.8V. Then

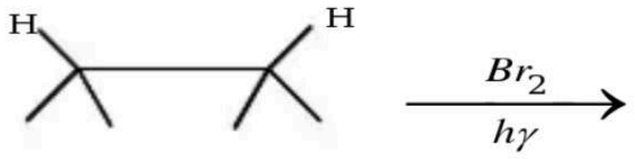
(1) Metal Q will displace the cation of P from its aqueous solution and deposit the metal P

(2) Both metals Q and R will displace the cation of P from its aqueous solution and deposit the metal P

(3) Metal P will displace the cation of R from its aqueous solution and deposit the metal R

(4) Metal R will displace the cation of Q and deposit the metal Q.

80. Major product of the following reaction is



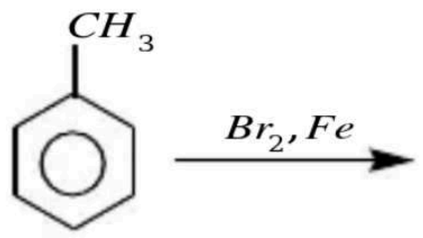
(1) 1 – Bromo -1, 1, 2, 2 – Tetra methyl ethane

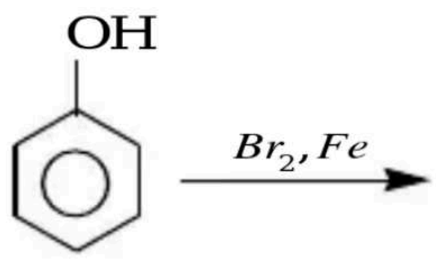
(2) 1 – Bromo – 2, 3, 3 – tri bromo propane

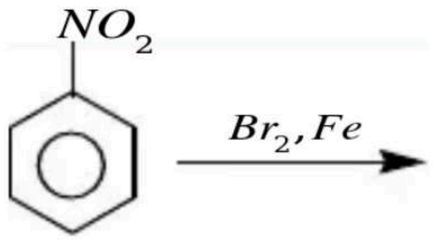
(3) 2 – bromo – 2, 3 – dimethyl butane

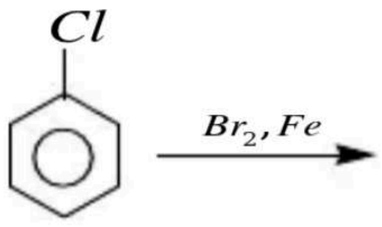
(4) 1 – bromo – 2, 3 – dimethyl butane

81. P – bromo benzene derivative can't be prepared from

(1) 

(2) 

(3) 

(4) 

82. Statment – I : Alkyl halides are not soluble in H_2O

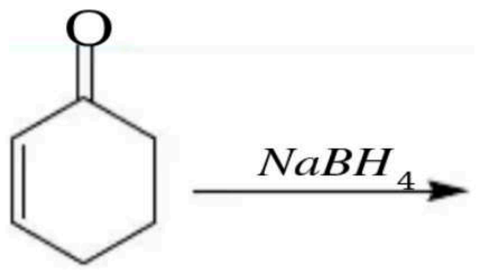
Statement – II : Alkyl halides does not form hydrogen bonds with H_2O although alkyl halide is polar in nature.

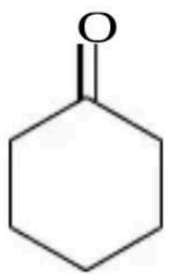
(1) Both S-I & S-II are incorrect
(2) S-I is correct but S-II is incorrect
(3) S-I is incorrect but S-II is correct
(4) Both S-I & S-II are correct

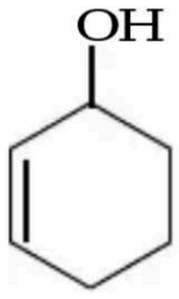
83. Assertion (A) : Carbocation formed in SN^2 reaction can rearrange.

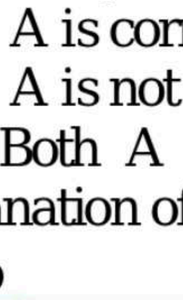
Reason (R) : Rearrangement of crabocation takes place to gain extra stability

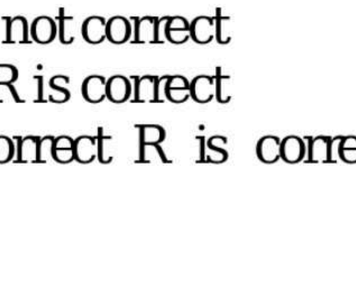
(1) Both A & R are correct but R is not correct explanation of A
(2) A is correct but R is not correct
(3) A is not correct but R is correct
(4) Both A & R are correct R is correct explanation of A

84.  Product formed in the above reaction is

(1) 

(2) 

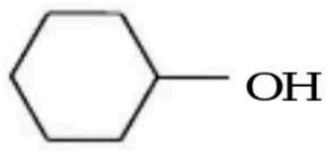
(3) 

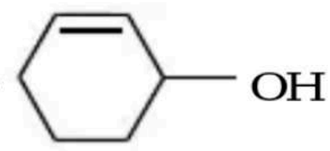
(4) 

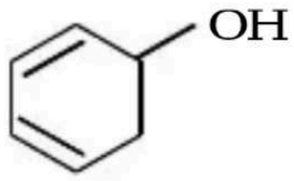
85. The most suitable reagent for the conversion of $R-CH_2-OH \rightarrow R-CHO$ is

(1) $KMnO_4$ (2) $K_2Cr_2O_7$
(3) CrO_3 (4) PCC

86. The correct order of ease of dehydration of the following alcohol is

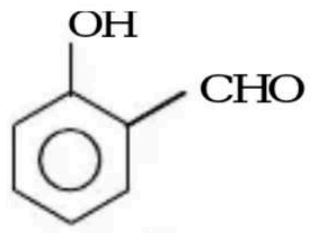
I. 

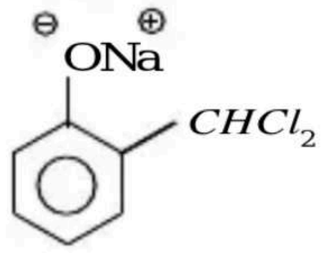
II. 

III. 

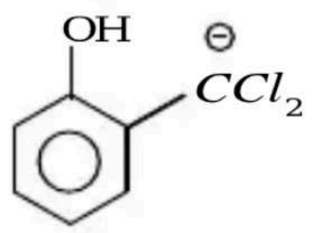
(1) I >II >III
(2) III >II >I
(3) I >III >II
(4) III >I >II

87. Which of the following is not formed in Reimer – Tie-mann reaction ?

(1) 

(2) 

(3)

(4) 

88. Reaction between acetone and methyl magnesium chloride followed by hydrolysis will give

(1) secondary butyl alcohol
(2) tert.butyl alcohol
(3) isobutyl alcohol
(4) isopropyl alcohol.

89. E_{1CB} reaction is given by which of the following

A. CF_3CHCl_2

B. $C_6H_5-CH\left(NO_2\right)-CHBr$

(1) A only (2) B only
(3) Neither A nor B (4) Both A & B

90. Crystalfield stabilization energy for high spin d^4 octahedral complex is

(1) $-1.8\Delta_0$ (2) $-1.6\Delta_0+P$
(3) $-1.2\Delta_0$ (4) $-0.6\Delta_0$